

Integrated Climate-intelligent Salt Production Ecosystem: A Multi-modal AI Framework For Sustainable Yield Optimization In Tropical Coastal Environments

Project ID: 25-26J-431

Supervised by: Dr. Nathali Silva
Co-supervised by: Mrs. Thilini Jayalath

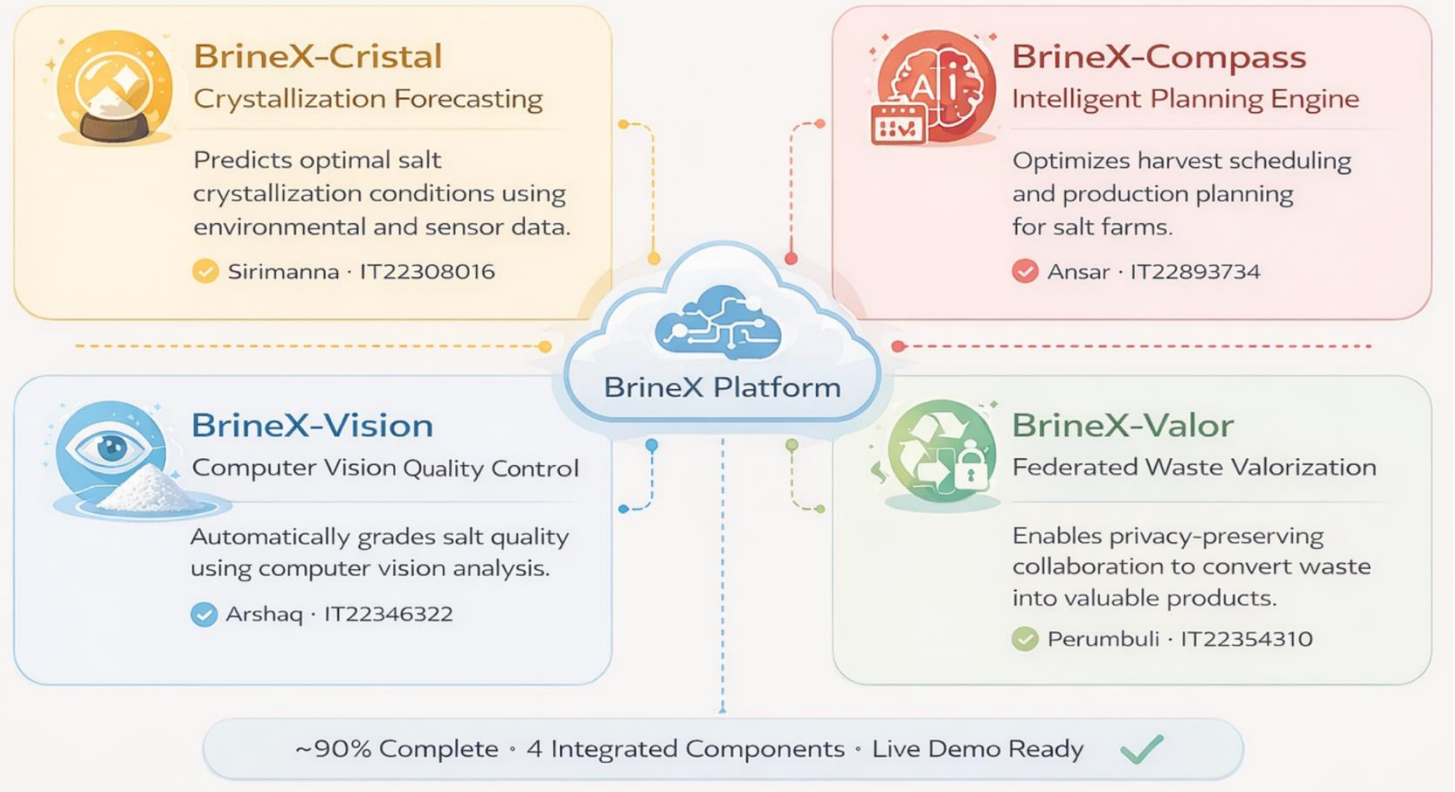
Integrated Climate-Intelligent Salt Production Ecosystem



Research Problem

Climate variability causes unpredictable salt yields, economic losses, and quality issues.

BrineX — An Integrated AI Platform for Salt Production



PP2 Comments and Actions Taken

Student	Comment	Action Taken
IT22893734 Ansar TD	Progressing well	Optimized frontend UX across Compass module User Acceptance Testing with PSS saltern operators
IT22308016 Sirimanna RDIB	Must be thorough when answering questions; improve communication skills	Conducted internal mock Q&A sessions Prepared clear technical explanations for all model decisions
IT22346322 Arshaq MJM	Added miscellaneous class for model training; needs quality attribute testing	Retrained the model, validated it through quality attribute testing, achieving overall F1 score of 99.28% (Precision 99.41%, Recall 99.14%) across all salt classes with no regression in existing detections.
IT22354310 Perumbuli PGRMD	Must explain how each factor (rainfall, humidity...) affects different waste types	Re-evaluated the baseline values used for environmental features, conducted a thorough literature review to verify the correctness of the physical models, and retrained the model

IT22308016 | SIRIMANNA R.D.I.B

Crystal - Crystallization Forecasting

Crystal predicts daily crystallization conditions and monthly salt production volumes using a dual-input Bidirectional LSTM and causal Linear Regression, giving landowners a 60-day planning horizon.

- Bidirectional LSTM trained on 1,095 daily records
- **Inputs:** 8 pond parameters + 14 weather features (OpenWeatherMap)
- **Output:** 60-day ahead forecast of all 8 crystallization parameters
- 532,832 trainable parameters; ONNX-deployed in NestJS microservice
- Test $R^2 = 0.8175$ | Validation $R^2 = 0.8805$ (no overfitting)
- Causal Linear Regression converts LSTM outputs $\rightarrow$ monthly production volumes
- Linear Regression $R^2 = 0.9732$ vs ARIMA $R^2 = 0.4100$
- Zero failed requests across 3,112 stress test invocations; avg response 88ms



IT22893734 | ANSAR T. D

Compass - Intelligent Planning Engine

Compass converts Crystal's production forecasts into market intelligence - demand estimates, two-month price forecasts, and ranked buyer recommendations - replacing informal price negotiation with data-driven decisions.

- Receives Crystal production forecasts via gRPC-to-HTTP bridge
- **Demand forecasting:** Seasonal yield ratio algorithm
- **Price forecasting:** SARIMAX
- **Seller recommendation:** Rule-based ranking by price_per_bag
- Trained on 36 months of PSS historical data
- 15 unit tests, 3 suites - zero failures; 13 endpoints validated



IT22346322 | ARSHAQ M.J.M

Vision - Computer Vision Quality Control System

Vision automates salt quality inspection using a YOLOv8m model that detects pure, impure, and unwanted crystals in real time, replacing subjective manual grading with objective, auditable quality scores.



- YOLOv8-medium trained on 1,952 custom-annotated images (Puttalam field data)
- 3-class detection: Pure, Impure, Unwanted foreign material
- **mAP50: 99.37%**
- **Precision: 99.41%**
- **Recall: 99.14%**
- HSV whiteness index per crystal; composite quality score per batch
- WebSocket real-time streaming to operator dashboard

IT22354310 | PERUMBULI P.G.R.M.D

Valor - Federated Learning for Waste Valorization

Valor predicts salt waste composition and valorisation potential across distributed salterns using a federated ensemble model, enabling collaborative mineral recovery planning without sharing sensitive operational data.

- Predicts 8 waste valorization outputs: solid waste (4), bittern (1), Epsom salt, Potash, Magnesium Oil
- Weighted ensemble: Gradient Boosting + Stacked Ensemble (XGBoost, LightGBM, RF, GBR) + Deep Neural Network
- Overall $R^2 = 0.9105$
- Physics-informed features: Evaporation Efficiency Index, Baumé proxy, brine density - account for 60% of ion predictive power
- Federated Learning (FedAvg) via AWS SQS: 0 bytes raw data shared; 98% of centralized performance (federated $R^2 = 0.904$)
- Converges within 50 communication rounds; tolerates 30% client dropout



Results and Testimonials

Component	Key Metric	Result
Crystal - LSTM	Test R ²	0.8175
Crystal - Linear Regression	R ² vs ARIMA	0.9732 vs 0.4100
Compass - Price (SARIMAX)	MAPE month+1 / month+2	1.16% / 1.35%
Vision - YOLOv8m	MAPE 0.5 / Precision / Recall	99.37% / 99.41% / 99.14%
Valor - Ensemble	Overall R ²	0.9907

Load Tests (Grafana k6)

Service	Status	Checks
Crystal	Production Ready	100%
Waste Valorisation	Production Ready	100%
Compass	Needs DB optimization	93.45%
Vision	Needs async queue	84.17%

UAT – User Acceptance Test

- UAT conducted with PSS stakeholders - solution fully accepted
- Stakeholders expressed strong satisfaction with the platform
- PSS to arrange a follow-up meeting with landowners and distributors to formally pitch BrineX for real world deployment

Limitations and Future Improvements

Limitations

- 36-month PSS dataset limits LSTM generalization
- Valor trained on synthetic thermodynamic data -needs real field measurements
- Vision works with one live camera feed at a time, and heavy inference can slow down the device.
- Trained on fixed salt grades and impurities; may misclassify unseen types or contaminants.

Future Improvements

- Replace manual daily pond measurements with real-time IoT sensor feeds
- Farm-specific yield ratios auto-fit as post-deployment transaction history builds
- Extend Vision to Hambantota, Mannar salterns
- Field validation with more active PSS landowners to measure real-world harvest decisions and economic impact.
- Add a user-driven self-learning pipeline to label new samples, retrain regularly, and expand impurity detection (e.g., sand, plastic, organic matter).



Thank you.

It's time for Q&A